

Erdős Problem #295

EP-295

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $N \geq 1$ and let $k(N)$ denote the smallest k such that there exist $N \leq n_1 < \dots < n_k$ with $1 = \frac{1}{n_1} + \dots + \frac{1}{n_k}$.
Is it true that $\lim_{N \rightarrow \infty} k(N) - (e - 1)N = \infty$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>